UrbanCart Analytics

Olist Dashboard Study Guide

Exact screenshots of all five Streamlit dashboards
with in-depth formulas and interpretation for every KPI and chart

Source: Brazilian E-Commerce Public Dataset by Olist (CC BY-NC-SA 4.0)
Window: Sep 2016 – Sep 2018 · Currency: BRL
Warehouse clock: analysis_date = MAX(order_date) = 2018-09-03
Person key: customer_unique_id (not Olist per-order customer_id)

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How to study this guide

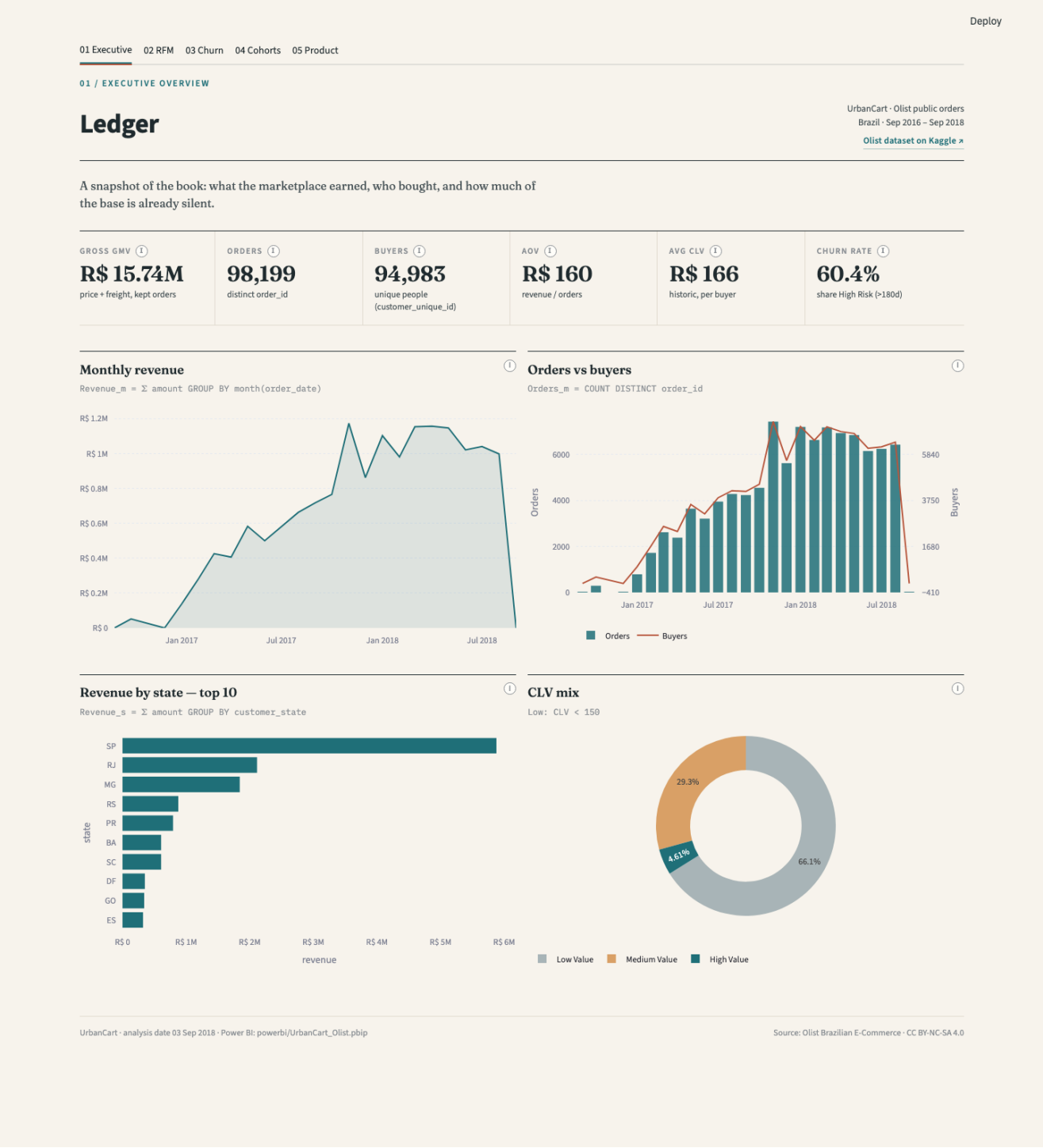
Each section opens with a full-page screenshot of the live Streamlit tab, then documents every KPI and chart: what it shows, the exact formula used in SQL/Python, and how to interpret it for business data mining. Recency and churn always use the dataset's last order date — never today's calendar — otherwise every 2018 buyer would look churned.

Core CLV identity used across Executive and CLV bands:

```
AOV_c = Σ amount_c / COUNT(orders)_c
lifespan_years_c = MAX(lifespan_days_c, 1) / 365
purchase_frequency_c = orders_c / lifespan_years_c
CLV_c = AOV_c × purchase_frequency_c × lifespan_years_c
```

01 — Executive Overview (Ledger)

Board-level health check for the Olist-backed UrbanCart book: how much merchandise was sold, how many people bought, what a typical order is worth, how valuable buyers look on a lifetime basis, and how much of the base has gone quiet. Every number uses kept orders only (canceled / unavailable dropped).



Screenshot · Streamlit tab “01 Executive” · UrbanCart Olist

01 — Executive Overview (Ledger) — detailed reference

KPIs

Gross GMV

```
GMV = Σ (price + freight_value) = Σ orders_fact.amount
```

Gross merchandise value across every kept order line. Amount is the line's mean unit price contribution plus freight allocated to that line. This is the top-line money metric for the extract.

Orders

```
Orders = COUNT(DISTINCT order_id)
```

Number of unique baskets. Multi-item orders still count as one order. Analysis date for the book is MAX(order_date) = 2018-09-03.

Buyers

```
Buyers = COUNT(DISTINCT customer_id)
```

People, not Olist's per-order customer_id. We map to customer_unique_id so RFM/CLV are not inflated.

AOV

```
AOV = GMV / Orders
```

Average order value for the whole book — revenue per basket.

Avg CLV

```
CLV = AOV_c × purchase_frequency_c × lifespan_years_c; then AVG(CLV)
```

Mean of historic customer CLV. Lifespan is floored at 1 day so a one-order buyer has CLV equal to that order (not zero).

Churn rate

```
ChurnRate = COUNT(High Risk) / COUNT(Buyers)
```

Share of buyers whose last order is more than 180 days before the dataset's last order date (not today's calendar).

Charts & tables

Monthly revenue (area)

```
Revenue_m = Σ amount GROUP BY DATE_TRUNC('month', order_date)
```

Time series of GMV by purchase month. Shows seasonality and growth through 2016–2018. Canceled/unavailable orders are already excluded upstream in transform_olist.py.

Orders vs buyers (dual axis)

```
Orders_m = COUNT DISTINCT order_id; Buyers_m = COUNT DISTINCT customer_id
```

Bars = order volume; line = distinct people that month. When orders rise faster than buyers, the same people are buying more often.

Revenue by state — top 10 (bar)

```
Revenue_s = Σ amount GROUP BY customer_state
```

Geographic concentration of GMV. State comes from the buyer's shipping state on their first kept order (dim_customers).

CLV mix (donut)

```
Low: CLV < 150; Medium: 150 ≤ CLV < 500; High: CLV ≥ 500 (BRL)
```

Share of buyers in each historic CLV band. Bands are tuned for Olist’s BRL order scale (not the old synthetic INR 3k/9k cuts).

02 — RFM Segmentation (Who still buys)

Behavioral segmentation on Recency, Frequency, and Monetary value. Recency and Monetary use NTILE(5). Frequency is rule-based (1 / 3 / 5) because ~97% of Olist buyers order only once — quintiles on frequency would collapse.

Deploy

01 Executive | 02 RFM | 03 Churn | 04 Cohorts | 05 Product

02 / RFM SEGMENTATION

Who still buys

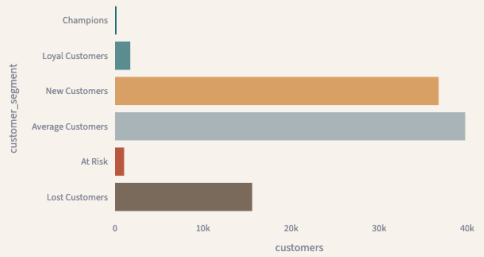
Recency · Frequency · Monetary
Scores 1–5, analysis date = last order
Olist dataset on Kaggle ↗

Quintiles on recency and spend. Frequency is rule-based: almost every Olist buyer purchases once, so quintiles on frequency would collapse.

CHAMPIONS	LOYAL CUSTOMERS	NEW CUSTOMERS	AVERAGE CUSTOMERS	AT RISK	LOST CUSTOMERS
124	1,726	36,753	39,770	1,037	15,573
R$ 69.5k GMV	R$ 512.6k GMV	R$ 6.06M GMV	R$ 7.92M GMV	R$ 308.2k GMV	R$ 867.7k GMV

Segment headcount

Champions: R=4, F=5, M=4



Revenue by segment

Revenue_seg = Σ monetary GROUP BY customer_segment



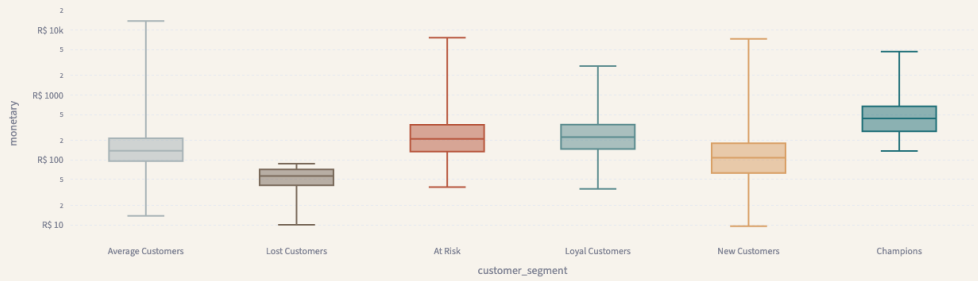
R × F heatmap

r_score = NTILE(5) ORDER BY recency DESC



Monetary spread by segment

monetary_c = Σ amount_c



Segment table

avg_recency = AVG(analysis_date − last_order)

Segment	Buyers	Avg recency (days)	Avg frequency	GMV (BRL)
Champions	124	96.4	3.53	69473.57
Loyal Customers	1726	139	2.03	512587.51
New Customers	36753	95.5	1	6058397.29
Average Customers	39770	319.1	1	7919167.74
At Risk	1037	387.5	2.08	308154.13
Lost Customers	15573	401.9	1	867746.79

UrbanCart · analysis date 03 Sep 2018 · Power BI: powerbi/UrbanCart_Olist.pbip

Source: Olist Brazilian E-Commerce · CC BY-NC-SA 4.0

Screenshot · Streamlit tab “02 RFM” · UrbanCart Olist

02 — RFM Segmentation (Who still buys) — detailed reference

KPIs

Segment headcount KPIs

```
R = analysis_date − last_order; F = COUNT DISTINCT orders; M = Σ amount
```

One KPI cell per RFM label with buyer count and GMV in that segment. Labels: Champions, Loyal, New, Average, At Risk, Lost.

Charts & tables

Segment headcount (horizontal bar)

```
Champions: R≥4, F=5, M≥4; Loyal: R≥3, F≥3; New: R≥4, F=1; At risk: R≤2, F≥3; Lost: R≤2, F=1, M≤2; else Average
```

How many people sit in each RFM label. Champions are recent, repeat, high-spend. New Customers are recent one-time buyers.

Revenue by segment (bar)

```
Revenue_seg = Σ monetary GROUP BY customer_segment
```

Where lifetime spend actually sits. A small Champions segment can still hold meaningful GMV.

R × F heatmap

```
r_score = NTILE(5) ORDER BY recency DESC; f_score = 1|3|5 by order count
```

Cell = buyer count for each (r_score, f_score). High R + high F is the sweet spot; high R + F=1 are recent one-timers.

Monetary spread by segment (box, log)

```
monetary_c = Σ amount_c
```

Distribution of lifetime spend inside each label. Log scale because a few Champions dwarf Average / New customers.

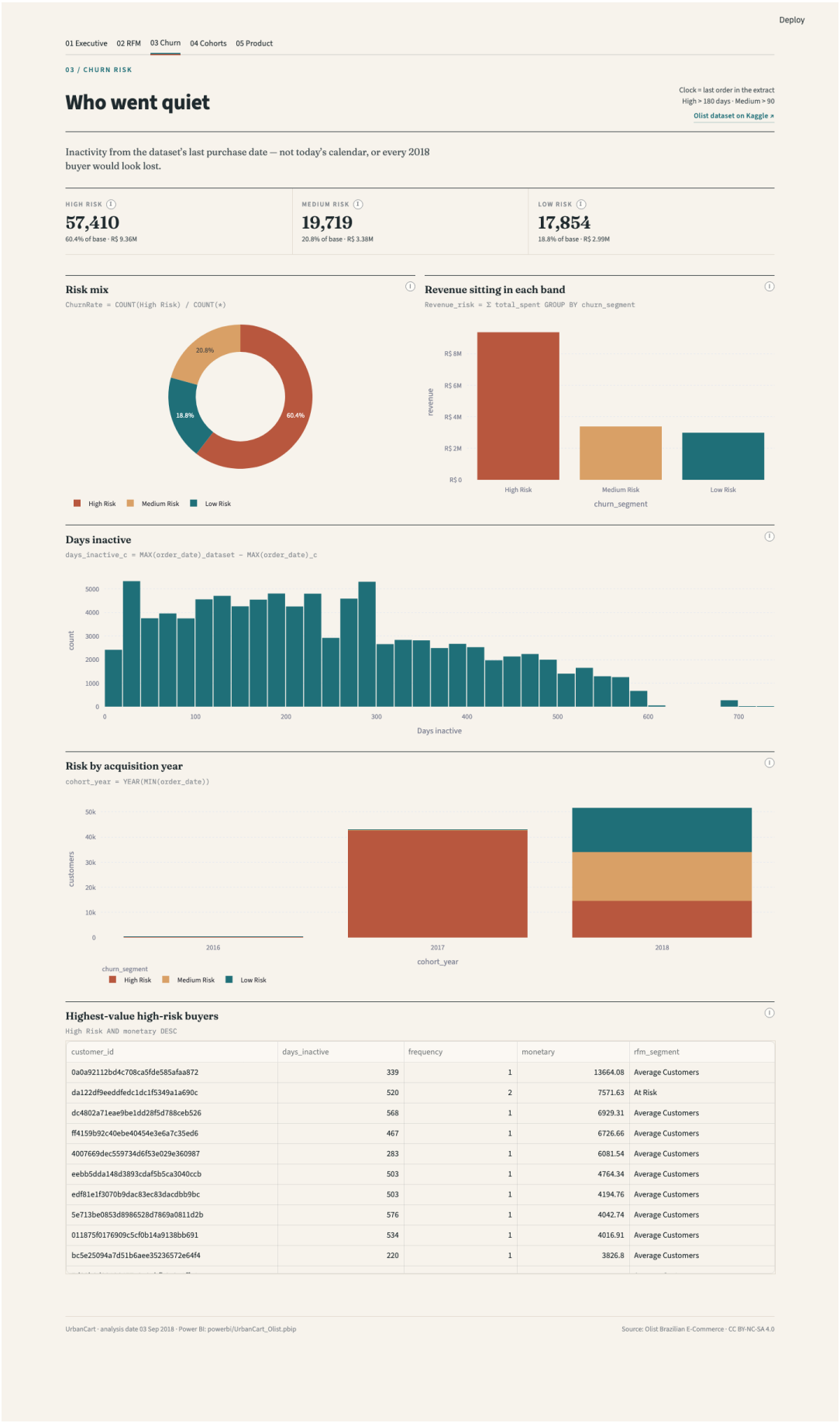
Segment table

```
avg_recency = AVG(R); avg_frequency = AVG(F); GMV = Σ M
```

Summary table: buyers, average days since last order, average order count, and total monetary per segment — for targeting briefs.

03 — Churn Risk (Who went quiet)

Inactivity-based churn proxy. Olist has no cancel/unsubscribe flag, so risk is defined from days since last purchase vs the extract’s last order date (2018-09-03).



Screenshot · Streamlit tab "03 Churn" · UrbanCart Olist

03 — Churn Risk (Who went quiet) — detailed reference

KPIs

High / Medium / Low Risk counts

```
days_inactive = analysis_date − last_order; High>180; Medium>90; else Low
```

Each KPI shows headcount, share of base, and lifetime GMV sitting in that band.

Charts & tables

Risk mix (donut)

```
ChurnRate = COUNT(High Risk) / COUNT(*)
```

Share of the base in each inactivity band. High Risk is the working definition of churn for this project.

Revenue sitting in each band (bar)

```
Revenue_risk = Σ total_spent GROUP BY churn_segment
```

Money at stake. High-risk revenue is lifetime GMV of people who have not ordered in >180 days — prioritise win-back here.

Days inactive (histogram)

```
days_inactive_c = MAX(order_date)_dataset − MAX(order_date)_c
```

Distribution of inactivity length. Long right tail = early buyers who never returned.

Risk by acquisition year (stacked)

```
cohort_year = YEAR(MIN(order_date)); COUNT by (year, churn_segment)
```

Older acquisition years have had more calendar time to go quiet, so High Risk share rises for earlier cohorts.

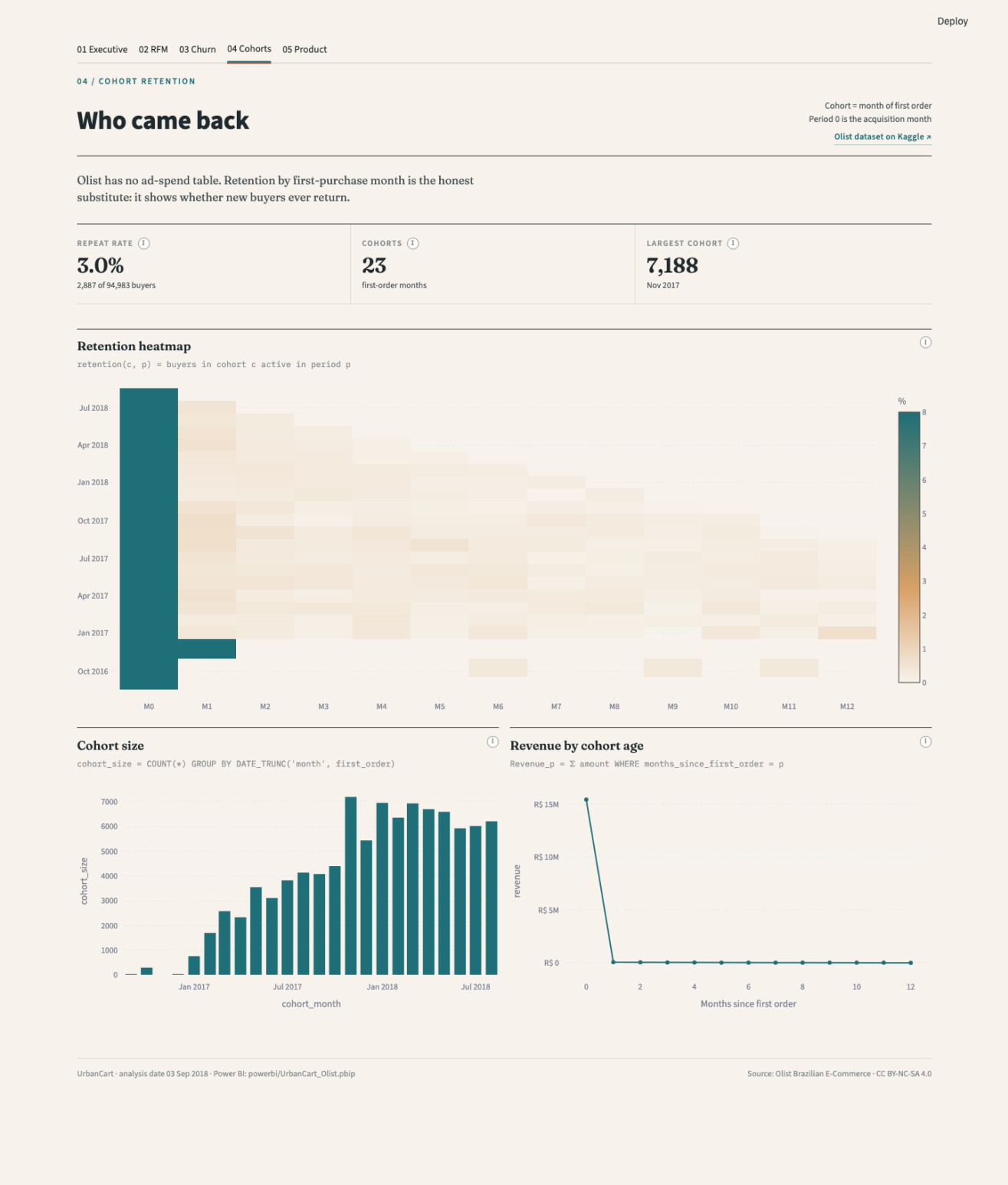
Highest-value high-risk buyers (table)

```
WHERE churn_segment = 'High Risk' ORDER BY monetary DESC
```

Target list: silent buyers ranked by lifetime spend, with RFM frequency and segment for outreach prioritisation.

04 — Cohort Retention (Who came back)

Replaces Marketing Attribution — Olist has no ad-spend table. Retention by first-purchase month is the honest view of whether new buyers ever return. Olist's true repeat rate is low; most shoppers buy once.



Screenshot · Streamlit tab "04 Cohorts" · UrbanCart Olist

04 — Cohort Retention (Who came back) — detailed reference

KPIs

Repeat rate

```
RepeatRate = COUNT(orders ≥ 2) / COUNT(buyers)
```

Share of people with at least two distinct orders in the extract.

Cohorts

```
cohort_month = DATE_TRUNC('month', MIN(order_date))
```

Number of first-order months with at least one kept buyer.

Largest cohort

```
cohort_size = COUNT(*) GROUP BY cohort_month
```

Headcount of the biggest acquisition month.

Charts & tables

Retention heatmap

```
retention(c,p) = active buyers in cohort c at period p / cohort_size_c
```

Rows = acquisition month; columns = months since first order (M0–M12). Cell = % of that cohort who ordered again in that later month. Colour scale capped near 8% so the faint repeat signal is visible. M0 is always ~100% (acquisition).

Cohort size (bar)

```
cohort_size = COUNT(*) GROUP BY first-order month
```

How many new buyers arrived each month.

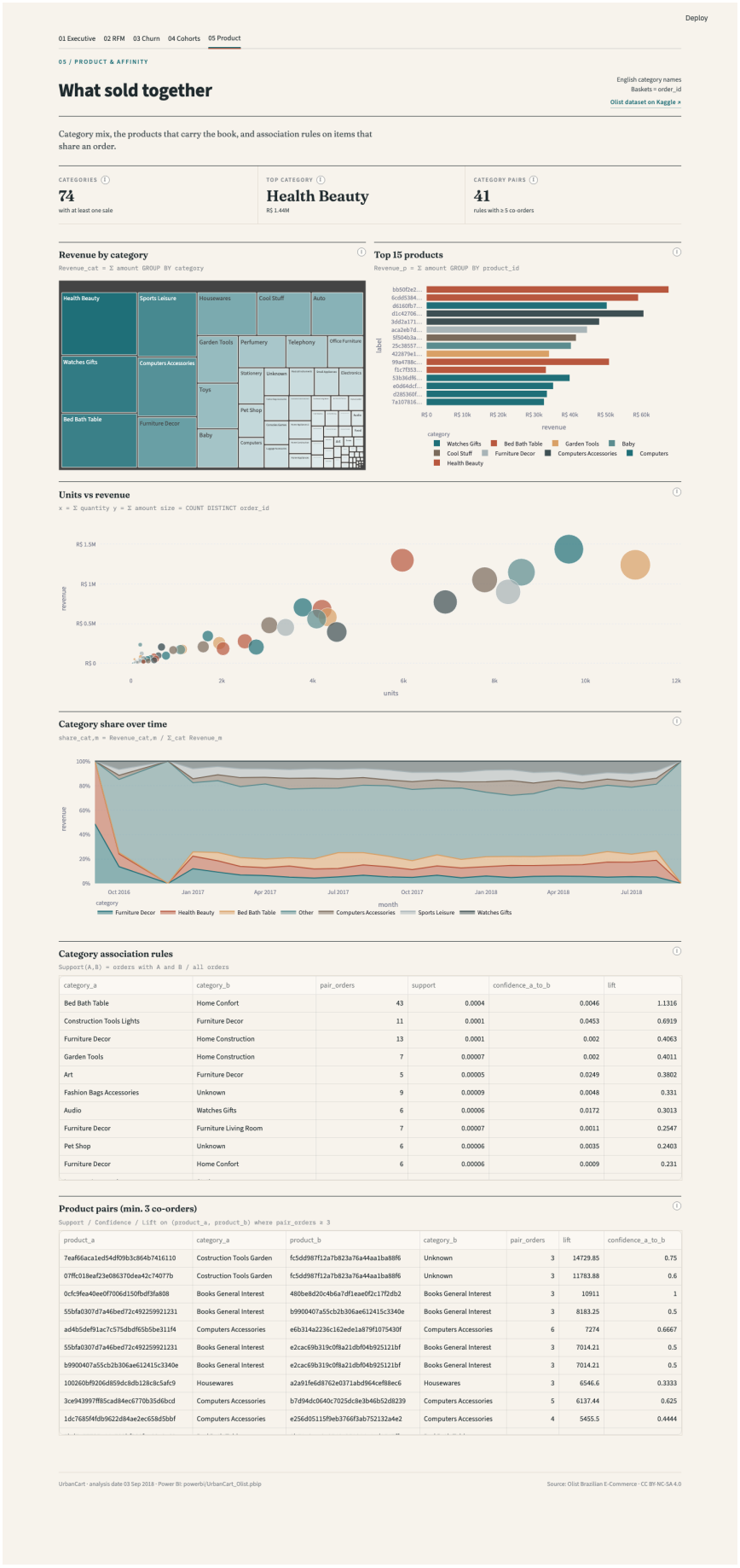
Revenue by cohort age (line)

```
Revenue_p = Σ amount WHERE months_since_first_order = p
```

All cohorts stacked: GMV booked in the nth month after first order. Period 0 is acquisition spend; later periods are repeat spend — usually much smaller on Olist.

05 — Product & Affinity (What sold together)

Category mix, top SKUs, and market-basket association rules. Basket grain is order_id. English category names come from Olist’s translation table.



category_a	category_b	pair_orders	support	confidence_a_to_b	lift
Bed Bath Table	Home Confort	43	0.0004	0.0046	1.1316
Construction Tools Lights	Furniture Decor	11	0.0001	0.0453	0.6919
Furniture Decor	Home Construction	13	0.0001	0.002	0.4063
Garden Tools	Home Construction	7	0.00007	0.002	0.4011
Art	Furniture Decor	5	0.00005	0.0249	0.3802
Fashion Bags Accessories	Unknown	9	0.00009	0.0048	0.331
Audio	Watches Gifts	6	0.00006	0.0172	0.3013
Furniture Decor	Furniture Living Room	7	0.00007	0.0011	0.2547
Pet Shop	Unknown	6	0.00006	0.0035	0.2403
Furniture Decor	Home Confort	6	0.00006	0.0009	0.231

product_a	category_a	product_b	category_b	pair_orders	lift	confidence_a_to_b
7eaf66aca1ed54df09b3c864b7416110	Costruction Tools Garden	fc5dd987f12a7b823a76a44aa1ba88f6	Unknown	3	14729.85	0.75
07ffc018eaf23e086370dea42c74077b	Costruction Tools Garden	fc5dd987f12a7b823a76a44aa1ba88f6	Unknown	3	11783.88	0.6
0cfc9fea40ee0f7006d150fbdf3fa808	Books General Interest	480be8d20c4b6a7df1eae0f2c17f2db2	Books General Interest	3	10911	1
55bfa0307d7a46bed72c49225921231	Books General Interest	b9900407a55cb2b306ae612415c3340e	Books General Interest	3	8183.25	0.5
ad4b5def91ac7c575dbdf65b5be311f4	Computers Accessories	e6b314a2236c162ede1a879f1075430f	Computers Accessories	6	7274	0.6667
55bfa0307d7a46bed72c49225921231	Books General Interest	e2cac69b319c0f8a21dbf04b925121bf	Books General Interest	3	7014.21	0.5
b9900407a55cb2b306ae612415c3340e	Books General Interest	e2cac69b319c0f8a21dbf04b925121bf	Books General Interest	3	7014.21	0.5
100260bf9206dd859dc8db128c8c5afc9	Housewares	a2a91fe6d8762e0371abd964cef88ec6	Housewares	3	6546.6	0.3333
3ce943997ff85cad84ec6770b35d6bcd	Computers Accessories	b7d94dc0640c7025dc8e3b46b52d8239	Computers Accessories	5	6137.44	0.625
1dc7685f4fdb9622d84ae2ec658d5bbf	Computers Accessories	e256d05115f9eb3766f3ab752132a4e2	Computers Accessories	4	5455.5	0.4444

Screenshot · Streamlit tab “05 Product” · UrbanCart Olist

05 — Product & Affinity (What sold together) — detailed reference

KPIs

Categories / Top category / Category pairs

```
COUNT DISTINCT category; argmax Σ amount; pair_orders ≥ 5
```

Catalog breadth, highest-GMV category, and how many category pairs co-occur in ≥5 baskets.

Charts & tables

Revenue by category (treemap)

```
Revenue_cat = Σ amount GROUP BY category
```

Area ∝ GMV. Quickly shows which English categories carry the book.

Top 15 products (bar)

```
Revenue_p = Σ amount GROUP BY product_id
```

Highest-GMV product ids. Olist does not publish product titles — the id is the analysis grain.

Units vs revenue (scatter)

```
x = Σ quantity; y = Σ amount; size = COUNT DISTINCT order_id
```

Each point is a category. High units + low revenue = cheap volume; high revenue + fewer units = premium mix.

Category share over time (100% area)

```
share_cat,m = Revenue_cat,m / Σ_cat Revenue_m
```

Top six categories by total GMV plus Other. Stack sums to 100% of that month’s revenue — mix shift over time.

Category association rules (table)

```
Support(A,B)=pair/all; Confidence(A→B)=pair/orders_with_A; Lift=Support/(SupportA×SupportB)
```

Market basket at category grain. Lift > 1 means the pair appears together more than chance. Most Olist baskets are single-category, so lift often sits near or below 1 — still useful for assortment.

Product pairs (table)

```
Same Support / Confidence / Lift on product_id where pair_orders ≥ 3
```

Sparse at SKU grain; short high-signal list for concrete cross-sell.

Quick formula cheat sheet

Topic	Formula
GMV	Σ (price + freight) on kept lines
AOV	GMV / DISTINCT orders
CLV	AOV × (orders/lifespan_years) × lifespan_years
RFM R	analysis_date – last_order (days)
RFM F	1 if 1 order; 3 if 2; 5 if ≥3
RFM M	Σ amount
Churn High	days_inactive > 180
Retention	active_in_period_p / cohort_size
Support(A,B)	orders with A∧B / all orders
Confidence(A→B)	orders with A∧B / orders with A
Lift(A,B)	Support(A,B) / (Support(A)×Support(B))

Repo: github.com/kavyarathod05/urban-cart-customer-analytics-clv · Live app: streamlit run app.py · Power BI: powerbi/UrbanCart_Olist.pbip